

RMDL Assessment 2: Data Analysis Report

B281160

1 1. Background

The Stroop test has become one of the most popular psychological tests to evaluate language control and the scope of the so-called bilingual advantage in the developmental linguistics literature. In summary (see MacLeod & MacDonald (2000) for an effective review) the test captures the subtle processing delay that occurs when we process competing and conflicting stimuli, most frequently realised as a word-colour test where participants read a word and respond with the font colour. Specifically, the Stroop test allows researchers to assess subtle differences in inhibition control and executive functions during automatic tasks, such as reading, when assessing the extent of bilingual lexical access. Despite being used extensively in bilingual acquisition research for decades, the Stroop task remains relevant as ever, especially given substantial new research findings relating to language non-selectivity (Kroll et al. 2012; Blumenfeld and Marian 2007). One well-established model for language non-selectivity in bilinguals and multilinguals is The Bilingual Interactive Activation model (BIA+) (Dijkstra and Van Heuven 2002) which posits that a bilingual individual's L1 and L2 is simultaneously activated due to an integrated L1 and L2 lexicon. Bilingual speakers must therefore suppress non-target languages (Linck, Hoshino, and Kroll 2008) to prevent cross linguistic interference from this shared lexicon. Consequently, the Stroop task remains an effective method of testing inhibitory control, automaticity, and cross-linguistic interference as a means of understanding non-selective activation in greater depth. This, rather than the notion of bilingual advantage, will be the broad focus of this study.

Much of this research has focussed on the role of age of acquisition (Bugg et al. 2007), proficiency (M giste 1985) and direction of language switching (Liu et al. 2019) on a bilingual's ability to suppress information from the non-target language. Despite remaining a contentious issue, the role of orthographic similarity is discussed far less frequently in the literature. According to the BIA+, orthographic similarity between languages may modulate the degree of cross linguistic activation by determining the amount of cross-linguistic bottom-up activation (Dijkstra and Van Heuven 2002; Coderre and Van Heuven 2014). Specifically, logographic languages appear to activate language centres in the brain bilaterally where alphabetic languages more predominantly activate left-hemisphere language systems, and thus cross-script language

studies provide an opportunity to better understand the neural representations of lexical processing (Coderre and Van Heuven 2014) and how script type modulates non-selective language activation.

However, previous Stroop experiments investigating the impact of cross-script language interference yield varying results. While some research indicates that script differences may provide stronger cues to suppress cross-linguistic interference, resulting in a smaller Stroop effect for different-script bilinguals (Coderre and Van Heuven 2014; Bosworth et al. 2021) other studies maintain that orthographic differences have no systematic relationship with the size of the Stroop effect (Lee et al. 1992) and dispute the validity of Van Heuven & Coderre (2014)' (2014)'s findings (Paap et al. 2015). Paap et al. (2015) note that the study – and the debate as a whole – suffers from no systematic way of measuring orthographical and phonological language similarity, nor the complex ways these two factors interact. Although designing a way to objectively measure these qualities is beyond the scope of the current paper, it highlights that more work is needed to address the ambiguous conclusions regarding orthographic similarity in cross-linguistic interference.

Disentangling the effects of orthographical similarity and phonological similarity is complex, and few natural languages provide the opportunity to explore the effect of orthographic similarity alone. One possible solution is to test bilingual speakers of languages which use more than one script, and explore whether there is an interference effect based on the script alone. Hindi-English bilinguals provide a unique opportunity to explore this problem; it is also relatively understudied in the existing literature, particularly with regards to the effect of orthography on non-selective activation. Hindi is typically written in the alpha-syllabary Devanagari script and shares no phonological system or cognates with English (Mishra and Singh 2014). However, it is increasingly common in the digital age for Hindi speakers to read and write online using Latin script (Wolf-Sonkin et al. 2019) for convenience. In this scenario, the Latin Hindi word is phonologically and lexically identical, but the use of Latin script creates an orthological overlap with a Latin alphabet language, such as English. In Hindi-English bilinguals, this potentially poses an issue where the retrieval cue determined by the script, as proposed by the BIA+ account (Dijkstra and Van Heuven 2002), is no longer a reliable lexical cue. It is therefore possible that Hindi-English bilinguals experience a greater Stroop effect in Latin script Hindi.

NB: please note that this background draws from work in B281160 RMDL Assignment 1.

1.1 1.1 Research Questions/Hypotheses

To this end, Hindi provides a unique opportunity to assess the effect of orthographic differences in inhibition control and lexical access in a within-language design. Although there are a small number of tests assessing the Stroop effect in Hindi-English bilinguals (Datta, Nebhinani, and Dixit 2019) there is no precedent in the previous literature assessing the Stroop effect in different scripts of the same language. This experiment therefore presents a novel opportunity

to explore how inhibitory control is modulated by orthography in an underrepresented area of the previous research, and to understand the effect of orthography in cross-script bilinguals where the phonology is held constant.

Subsequently, the present study asks the following research questions:

1. Do Hindi-English bilinguals exhibit a stronger Stroop interference effect (i.e., the form of longer reaction times) when reading Latin script Hindi words versus reading Devanagari Hindi words?
2. What is the difference in the magnitude of interference (in the form of reaction times) between Devanagari Hindi and English conditions, Latin Hindi and English conditions, and Devanagari Hindi and Latin Hindi conditions?

In line with the BIA+ model, we hypothesise that the greatest Stroop effect will be found in the Latin script Hindi conditions as the script will no longer be an effective retrieval cue for non-target language suppression. Subsequently, this would also suggest that the Latin Hindi vs Devanagari Hindi conditions, where the phonology between trials is identical but the script is different, will show the largest degree of interference between all 3 condition comparisons as the phonological representations will compete with the orthography.

2 Methodology

2.1 Participants

The study sample consists of 5 participants between 18 and 40 years of age; please see Table 1 for an overview of participant characteristics. The sample group was smaller than initially planned due to unexpected constraints with participant recruitment. The participants are highly proficient sequential bilingual speakers of Hindi and English, although interestingly there was considerable variation in Hindi reading proficiency. All participants acquired Hindi at home before the age of 5 and acquired English at school from the age of 5 onwards. The participants did not disclose any clinical history of colour blindness, head trauma, neurological disease, substance abuse, or sedative use in the 72 hours prior to taking the experiment. This was to eliminate any neurological conditions that would interfere with the participants' reaction times.

Table 1: *Table 1: Participant language characteristics, including Age of Acquisition (AoA) and spoken/reading proficiency data*

	Mean AoA (years)	Mean spoken proficiency (/10)	Mean reading proficiency (/10)
English	3	9.3	9.3
Hindi	1.66	8.6	5.3

2.2 2.2 Materials

Participants completed an online version of the traditional colour-word Stroop task in Hindi, English, and Latin script Hindi conducted through Testable (Rezlescu, 2020). The stimuli words consisted of red, yellow, blue and green in each language condition, as seen in Table 2 below. Participants responded using the up, down, left, and right keys on the keyboard. In alignment with Datta et al. (2019) only four colours were chosen.

Table 2: *Table 2: Stroop stimuli by language condition*

Hindi	English	Latin script Hindi
	Red	Laal
	Yellow	Peela
	Green	Hara
	Blue	Neela

Table 3: *Table 3: Congruent and incongruent materials for each language condition*

Hindi congruent	Hindi incongruent	English congruent	English incongruent	Latin Script Hindi Congruent	Latin Script Hindi Incongruent
		RED	RED	LAAL	LAAL
		YELLOW	YELLOW	PEELA	PEELA
		GREEN	GREEN	HARA	HARA
		BLUE	BLUE	NEELA	NEELA

2.3 2.3 Procedure

Prior to beginning the experimental tasks, participants first completed a consent form informing them of any associated risks or ethical considerations. This was followed by the LEAP-Q questionnaire to confirm their proficiency and age of acquisition of both languages. Participants were emailed a link to the experiment on Testable, with a randomised participant code for anonymity, and invited to complete the test at their convenience in a quiet space where they can concentrate uninterrupted for the duration of the test.

The experiment began with instructions in English explaining the basic premise of the Stroop task: colour words in English, Hindi, and Latin script Hindi will appear in the middle of the screen and participants should respond with the colour of the font, not the colour represented by the word, as quickly as possible using LEFT, RIGHT, UP and DOWN on their keyboard to represent RED, YELLOW, GREEN and BLUE respectively. To familiarise themselves with the process, they completed a short training phase of 10 items where they respond with the font

colour of a non-lexical item “XXXXX” and receive written feedback confirming their accuracy. The accuracy and reaction time scores for the training phase were not assessed. They then progressed to the experimental Stroop tasks.

These were divided into three main experimental blocks. The first experimental block assesses the Stroop effect in English, followed by the second in Hindi, and finally the third in Latin script Hindi. In each block, there were 12 congruent items, where the font colour matches the word, and 12 incongruent items, where the font colour does not match the word. Please see Table 2 for an example of congruent versus incongruent conditions. Following the instructions in the training phase, participants were instructed to respond on their keyboard with the colour of the font and not the colour represented by the word on screen, with LEFT for red, RIGHT for yellow, UP for green and DOWN for blue. The 72 experimental items in each language condition will be pseudo-randomised to ensure the same colours and words do not follow consecutively. There will be a 5-minute break between each experiment. The Stroop task lasted no more than 5 minutes. The key outcome of interest for this study are reaction times, recorded in milliseconds (ms) for each participant and each item across all three experimental blocks. Accuracy is a secondary variable of interest for the English and Latin script Hindi conditions.

2.4 2.4 Statistical analysis

A Bayesian Lognormal regression analysis will be run in R using the brms program (v4.1.2; R Core Team 2021) to assess the relationship between reaction times, language condition, and congruent or incongruent conditions. Incongruent conditions eliciting the Stroop effect are of particular interest, but the analysis includes congruent conditions to explore the effect of orthography across language conditions more globally.

This is an entirely within-participant experimental design to mitigate for between group effects, therefore it is essential to fit a varying effects model which will control for individual variation in each experimental trial.

$RT \sim \text{language} + \text{Trialtype} + (\text{language} * \text{Trialtype} \mid \text{participant_ID})$

This structure models reaction times (RT) as a dependent variable of trial language and congruency. As all participants were exposed to the same trials, language and trial type were included as varying slopes.

Regarding the research questions, the statistical analysis will be used as follows:

1. *“Do Hindi-English bilinguals exhibit a stronger Stroop interference effect (i.e., the form of longer reaction times) when reading Latin script Hindi words versus reading Devanagari Hindi words?”*

Reaction times in the **incongruent** Latin script Hindi conditions and the Devanagari Hindi will first be compared to determine whether the use of Latin script creates orthological interference

when phonology and lexical content of the words are identical. Key summary measures (i.e., mean, median, mode, sd) of the raw data and the regression data will be calculated to compare these.

“2. What is the difference in the magnitude of interference (in the form of reaction times) between Devanagari Hindi and English conditions, Latin Hindi and English conditions, and Devanagari Hindi and Latin Hindi conditions?”

Summary measures of reaction times (as described above) will be compared between congruent and incongruent trials across all language conditions to establish the largest within-language Stroop effect. We will then calculate the probability of difference between incongruent trials in Devanagari and Latin Hindi, Latin Hindi and English, and English and Devanagari Hindi to establish the magnitude of interference.

2.5 2.5 Open Research Statement

This project is fully committed to Open Research practices.

All project items and relevant data are available through the below GitHub repository:
https://github.com/sa-lawrence/RMDL2_B281160

Anonymised participant data from Testable can be found under /results/. A .csv version of the Testable experiment is available under /materials/ along with the LEAP-Q questionnaire for language experience and proficiency.

3 3. Results

3.1 3.1 Workspace set-up

```
library(tidyverse)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v dplyr      1.1.4      v readr      2.1.5
v forcats    1.0.1      v stringr    1.5.2
v ggplot2    4.0.0      v tibble     3.3.0
v lubridate  1.9.4      v tidyr      1.3.1
v purrr      1.1.0
-- Conflicts ----- tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

```
library(brms)
```

```
Loading required package: Rcpp  
Loading 'brms' package (version 2.23.0). Useful instructions  
can be found by typing help('brms'). A more detailed introduction  
to the package is available through vignette('brms_overview').
```

```
Attaching package: 'brms'
```

```
The following object is masked from 'package:stats':
```

```
ar
```

```
library(tidybayes)
```

```
Attaching package: 'tidybayes'
```

```
The following objects are masked from 'package:brms':
```

```
dstudent_t, pstudent_t, qstudent_t, rstudent_t
```

```
library(ggdist)
```

```
Attaching package: 'ggdist'
```

```
The following objects are masked from 'package:brms':
```

```
dstudent_t, pstudent_t, qstudent_t, rstudent_t
```

```
library(posterior)
```

```
This is posterior version 1.6.1
```

```
Attaching package: 'posterior'
```

```
The following objects are masked from 'package:stats':
```

```
mad, sd, var
```

The following objects are masked from 'package:base':

```
%in%, match
```

```
library(bayesplot)
```

This is bayesplot version 1.14.0

- Online documentation and vignettes at mc-stan.org/bayesplot
- bayesplot theme set to bayesplot::theme_default()
 - * Does `_not_` affect other ggplot2 plots
 - * See `?bayesplot_theme_set` for details on theme setting

Attaching package: 'bayesplot'

The following object is masked from 'package:posterior':

```
rhat
```

The following object is masked from 'package:brms':

```
rhat
```

```
library(tinytex)
```

Warning: package 'tinytex' was built under R version 4.5.2

3.2 3.2 Data preparation

First, the Testable files will be read and combined into a single tibble for processing. Note that the first 3 rows of each raw data file were eliminated to remove Testable instructions for cleaner analysis.

```
files_to_read = list.files(  
  path = "./Data/",  
  pattern = "*.csv$",  
  recursive = TRUE,  
  full.names = TRUE  
)
```

```
data_lst <- lapply(files_to_read, function(x) {
  read_csv(x, skip = 3) |>
  mutate(
    participant_ID = response[1]
  )
})
```

Rows: 81 Columns: 24

-- Column specification -----

Delimiter: ","

chr (14): type, stim1, textColor, condition1, condition2, stimFormat, key, f...

dbl (9): rowNo, ITI, random, ITI_ms, ITI_f, ITI_fDuration, timestamp, RT, c...

lgl (1): responseCode

i Use `spec()` to retrieve the full column specification for this data.

i Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

Rows: 81 Columns: 24

-- Column specification -----

Delimiter: ","

chr (14): type, stim1, textColor, condition1, condition2, stimFormat, key, f...

dbl (9): rowNo, ITI, random, ITI_ms, ITI_f, ITI_fDuration, timestamp, RT, c...

lgl (1): responseCode

i Use `spec()` to retrieve the full column specification for this data.

i Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

Rows: 81 Columns: 24

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chr (14): type, stim1, textColor, condition1, condition2, stimFormat, key, f...

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lgl (1): responseCode

i Use `spec()` to retrieve the full column specification for this data.

i Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

Rows: 81 Columns: 24

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chr (14): type, stim1, textColor, condition1, condition2, stimFormat, key, f...

dbl (9): rowNo, ITI, random, ITI_ms, ITI_f, ITI_fDuration, timestamp, RT, c...

lgl (1): responseCode

```

i Use `spec()` to retrieve the full column specification for this data.
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Rows: 81 Columns: 24
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Delimiter: ","
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lgl (1): responseCode

```

```

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
Rows: 81 Columns: 24
-- Column specification -----
Delimiter: ","
chr (14): type, stim1, textColor, condition1, condition2, stimFormat, key, f...
dbl (9): rowNo, ITI, random, ITI_ms, ITI_f, ITI_fDuration, timestamp, RT, c...
lgl (1): responseCode

```

```

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
Rows: 81 Columns: 24
-- Column specification -----
Delimiter: ","
chr (14): type, stim1, textColor, condition1, condition2, stimFormat, key, f...
dbl (9): rowNo, ITI, random, ITI_ms, ITI_f, ITI_fDuration, timestamp, RT, c...
lgl (1): responseCode

```

```

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
Rows: 81 Columns: 24
-- Column specification -----
Delimiter: ","
chr (14): type, stim1, textColor, condition1, condition2, stimFormat, key, f...
dbl (9): rowNo, ITI, random, ITI_ms, ITI_f, ITI_fDuration, timestamp, RT, c...
lgl (1): responseCode

```

```

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
Rows: 81 Columns: 24
-- Column specification -----
Delimiter: ","
chr (14): type, stim1, textColor, condition1, condition2, stimFormat, key, f...
dbl (9): rowNo, ITI, random, ITI_ms, ITI_f, ITI_fDuration, timestamp, RT, c...

```

```

lgl (1): responseCode

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
Rows: 81 Columns: 24
-- Column specification -----
Delimiter: ","
chr (14): type, stim1, textColor, condition1, condition2, stimFormat, key, f...
dbl (9): rowNo, ITI, random, ITI_ms, ITI_f, ITI_fDuration, timestamp, RT, c...
lgl (1): responseCode

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.

```

```
full_RMDLSTROOP <- bind_rows(data_lst)
```

Next, the data was filtered to remove filler conditions from the testing phase and condition columns were renamed for clarity. All relevant columns were then selected and processed as a new tibble (“Stroop_filter”) which will be used to calculate the raw data summary measures.

```

#NB: Trialtype includes congruent and incongruent trials. Congruent condition is where the f

Stroop_filter <- full_RMDLSTROOP |>
  filter(condition1 %in% c("congruent", "incongruent"))

Stroop_filter <- Stroop_filter |>
  rename(Trialtype = condition1, language = condition2)

Stroop_filter <- Stroop_filter |>
  select(rowNo, type, stim1, textColor, Trialtype, language, key, response, RT,
         correct, participant_ID)

```

3.3 3.3 Raw data summary measures

As stated in Section 2.4, the key summary measures of interest are the mean, median, standard deviation (SD), and minimum/maximum ranges of the participants’ reaction times, grouped by trial type and language.

```

#Calculating summary measures.
Stroop_filter |>
  group_by(language, Trialtype) |>

```

```
summarise(
  mean = mean(RT), sd = sd(RT), median(RT), min(RT), max(RT)) |>
knitr::kable(digits = 2, col.names = c("Trial Language", "Congruency
                                         Condition", "Mean RT", "SD RT",
                                         "Median RT", "Min RT", "Max RT"),
  caption = "Table 4: Summary measures of reaction times (RT) across language and congruency
```

`summarise()` has grouped output by 'language'. You can override using the `.groups` argument.

Table: Table 4: Summary measures of reaction times (RT) across language and congruency conditions

Trial Language	Congruency Condition	Mean RT	SD RT	Median RT	Min RT	Max RT
english	congruent	1559.00	1242.48	1225.5	463	8877
english	incongruent	1811.48	1144.27	1430.5	480	5690
hindi	congruent	1281.68	796.94	1080.0	500	4590
hindi	incongruent	1820.48	1291.75	1300.5	488	5954
latin hindi	congruent	1253.12	852.46	941.0	441	4334
latin hindi	incongruent	1201.97	624.77	964.0	491	3476

Surprisingly, mean RTs in the Latin script Hindi condition were the fastest across incongruent and congruent trials. Within the congruent conditions, English had the highest mean RT at 1559ms followed by Devanagari Hindi at 1261.68ms and 1253.12ms for Latin Hindi. Within the incongruent conditions, Devanagari Hindi was the slowest at 1820.48ms, followed by English at 1811.48ms and Latin Hindi at 1201.97ms. This means the mean RTs for Latin script Hindi were actually faster in the incongruent trials than the congruent trials. Latin Hindi also had the lowest median RTs (congruent = 941ms, incongruent = 964ms) of all language conditions and the lowest SD (congruent = 852.46, incongruent = 624.77), the latter of which indicates less variance than the other two language conditions.

```
Stroop_filter |>
  group_by(participant_ID, language, Trialtype) |>
  summarise(
    prop_accuracy = mean(correct)*100) |>
knitr::kable(digits = 2, col.names = c("Participant ID", "Language",
                                         "Congruency", "Accuracy %"),
  caption = "Table 5: Summary measures of accuracy
across language and congruency conditions by participant")
```

`summarise()` has grouped output by 'participant_ID', 'language'. You can override using the `.groups` argument.

Table 4: Table 5: Summary measures of accuracy across language and congruency conditions by participant

Participant ID	Language	Congruency	Accuracy %
7138	english	congruent	100.00
7138	english	incongruent	88.89
7138	hindi	congruent	97.22
7138	hindi	incongruent	94.44
7138	latin hindi	congruent	100.00
7138	latin hindi	incongruent	100.00
7139	english	congruent	100.00
7139	english	incongruent	100.00
7139	hindi	congruent	91.67
7139	hindi	incongruent	100.00
7139	latin hindi	congruent	100.00
7139	latin hindi	incongruent	100.00
8747	english	congruent	100.00
8747	english	incongruent	100.00
8747	hindi	congruent	100.00
8747	hindi	incongruent	66.67
8747	latin hindi	congruent	100.00
8747	latin hindi	incongruent	91.67
8898	english	congruent	100.00
8898	english	incongruent	100.00
8898	hindi	congruent	100.00
8898	hindi	incongruent	100.00
8898	latin hindi	congruent	91.67
8898	latin hindi	incongruent	100.00
9565	english	congruent	100.00
9565	english	incongruent	100.00
9565	hindi	congruent	100.00
9565	hindi	incongruent	100.00
9565	latin hindi	congruent	100.00
9565	latin hindi	incongruent	100.00

```
Stroop_filter |>
  group_by(language, Trialtype) |>
  summarise(
    mean(correct)*100) |>
  knitr::kable(digits = 2, col.names = c("Language", "Congruency",
    "Mean accuracy %"),
```

```
caption = "Table 6: Summary measures of accuracy
across language and congruency conditions")
```

``summarise()`` has grouped output by 'language'. You can override using the ``groups`` argument.

Table 5: Table 6: Summary measures of accuracy across language and congruency conditions

Language	Congruency	Mean accuracy %
english	congruent	100.00
english	incongruent	96.67
hindi	congruent	98.33
hindi	incongruent	91.67
latin hindi	congruent	98.33
latin hindi	incongruent	98.33

Tables 5 and 6 show the accuracy of the participants' responses across language conditions and congruency, either by participant average (Table 5) or grouped by language and congruency condition (Table 6). These show that participants overall were the least accurate in the Devanagari Hindi incongruent condition, and most accurate in the congruent English conditions.

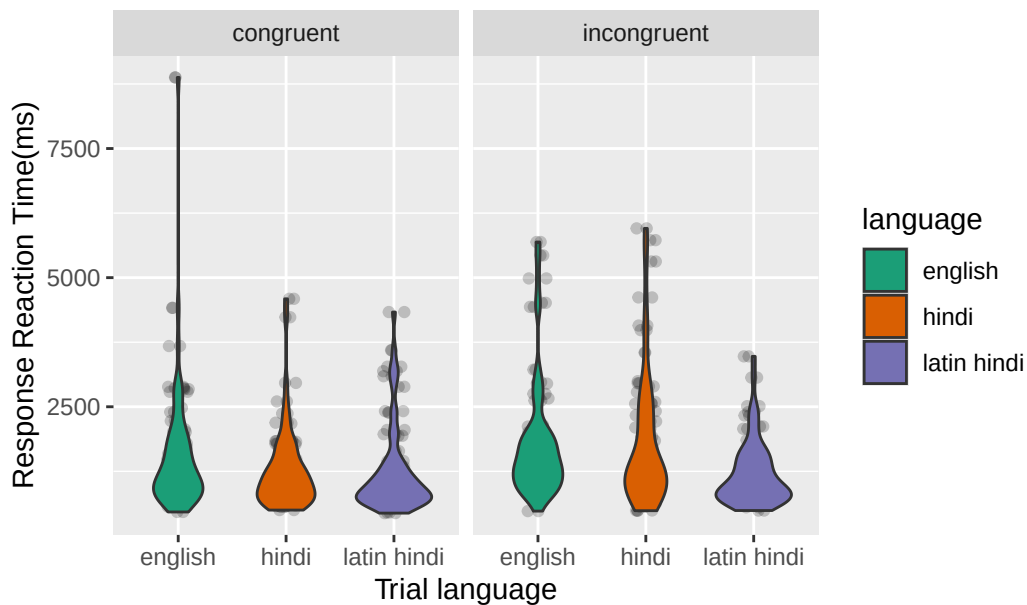
3.4 3.4 Plots: Raw data summaries

```
#Plotting the raw RT data.

jitterviolinRT_raw <- Stroop_filter |>
  ggplot(aes(language, RT, fill = language)) +
  geom_jitter(alpha = 0.2, width = 0.1) +
  geom_violin(width = 0.7) +
  scale_fill_brewer(palette = "Dark2") +
  facet_grid(cols = vars(Trialtype)) +
  labs(
    title = "Figure 1: Mean RTs by language and congruency",
    x = "Trial language",
    y = "Response Reaction Time(ms)"
  )

jitterviolinRT_raw
```

Figure 1: Mean RTs by language and congruency



This is a jitter plot of RTs grouped by language and separated by congruency condition in two panels. Within each panel, the participant's language is represented by colour, with green for English, orange for Hindi, and purple for Latin Hindi. The jitter plots show the individual raw data points, and the violin overlay indicates the density and variability in participants' responses. As shown in the summary conditions above, there appears to be greater variability in the RTs in incongruent conditions for English and Devanagari Hindi, with the exception of the outliers in the English congruent condition, reporting RTs of over 8000ms. Latin Hindi is the exception to this pattern, with similar density width patterns yet less overall variability in RT range.

```
#| label: raw-jitter-accuracy

accuracy_stroop_filter <- Stroop_filter |>
  group_by(participant_ID, Trialtype, language) |>
  summarise(
    prop_accuracy = round(mean(correct),2))
```

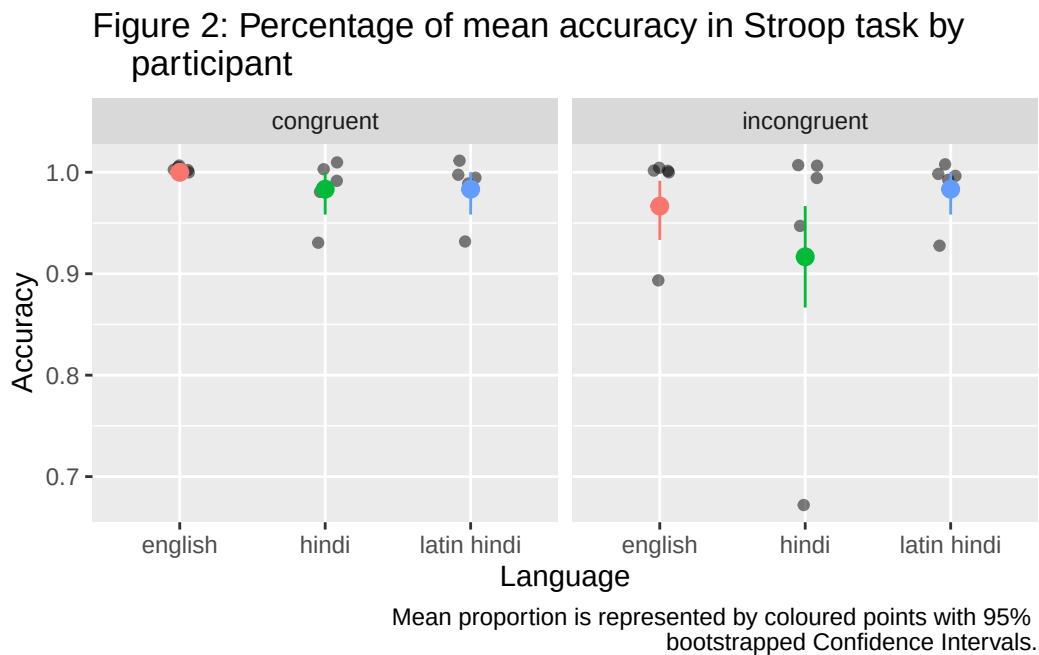
``summarise()`` has grouped output by 'participant_ID', 'Trialtype'. You can override using the ``.groups`` argument.

```
ggplot() +
  geom_jitter(
```

```

    data = accuracy_stroop_filter, aes(language, prop_accuracy),
    width = 0.1, alpha = 0.5
  ) +
  facet_grid(cols = vars(Trialtype)) +
  stat_summary(
    data = Stroop_filter,
    aes(language, correct, colour = language),
    fun.data = "mean_cl_boot", size = 0.5
  ) +
  labs(
    title = "Figure 2: Percentage of mean accuracy in Stroop task by
    participant",
    caption = "Mean proportion is represented by coloured points with 95%
    bootstrapped Confidence Intervals.",
    x = "Language",
    y = "Accuracy"
  ) +
  theme(legend.position = "none")

```



```

facet_grid(cols = vars(Trialtype))

```

<ggproto object: Class FacetGrid, Facet, gg>

```

attach_axes: function
attach_strips: function
compute_layout: function
draw_back: function
draw_front: function
draw_labels: function
draw_panel_content: function
draw_panels: function
finish_data: function
format_strip_labels: function
init_gtable: function
init_scales: function
map_data: function
params: list
set_panel_size: function
setup_data: function
setup_panel_params: function
setup_params: function
shrink: TRUE
train_scales: function
vars: function
super: <ggproto object: Class FacetGrid, Facet, gg>

```

Figure 2 is a jitter plot depicting the mean accuracy rating per participant by language and congruency condition. Mean accuracy scores were calculated per participant (as seen in section 3.3). Then, means of these means and confidence intervals were fitted with the `stat_summary()` function and plotted with `geom_jitter()`. As with Figure 1 above, congruency and incongruency are separated into 2 panels to assess the difference in Stroop effect from the incongruent conditions. English data is in red, Hindi is in green, and Latin Hindi is in blue. Accuracy between congruent and incongruent varied most in the English condition, while accuracy in Latin Hindi remains consistent between congruency conditions. Incongruent Hindi trials shows the greatest variability in accuracy, with a possible outlier to be considered.

4 4. Lognormal regression modelling

As stated in the statistical analysis section (2.4) this data will be analysed using a Bayesian Lognormal regression model using the `brms` function. This model was chosen for two key reasons: firstly, the lognormal distribution family is most appropriate for variables that are continuous and only positive or above 0, and is therefore well suited to analyse reaction times, and secondly, RTs tend to be right skewed as they are bound to positive numbers, so they need to be logged to approximate a Gaussian distribution.

This model is designed to assess the relationship between RTs, language script, and congruency conditions to see how the Stroop effect and longer reaction times are affected by either independent variable. Crucially, it was fitted with varying slopes to account for inter-participant variability. As this is a within-participant design, the within participant variables to include in the slopes are the same as the main model. Note that, as accuracy and proficiency are not of primary relevance to our research questions, these were not included in the models below.

Finally, the model was fitted with 0.999 adapt_delta and a max_treedepth of 15 to avoid divergent transitions.

```
#running the lognormal regression model

#| label: initial model

ENHIstroopbrm <- brm(
  RT ~ language * Trialtype + (language * Trialtype | participant_ID),
  family = lognormal,
  data = Stroop_filter,
  cores = 4,
  seed = 6789,
  file = "data/cache/ENHIstroopbrm8",
  control = list(adapt_delta = 0.999, max_treedepth = 15)
)
```

The summary measures are as follows below:

```
summary(ENHIstroopbrm)
```

```
Family: lognormal
Links: mu = identity
Formula: RT ~ language * Trialtype + (language * Trialtype | participant_ID)
Data: Stroop_filter (Number of observations: 720)
Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
       total post-warmup draws = 4000
```

Multilevel Hyperparameters:

~participant_ID (Number of levels: 5)

	Estimate
sd(Intercept)	0.37
sd(languagehindi)	0.26
sd(languagelatinhindi)	0.29
sd(Trialtypeincongruent)	0.32

sd(languagehindi:Trialttypeincongruent)	0.15
sd(languagelatinhindi:Trialttypeincongruent)	0.25
cor(Intercept,languagehindi)	-
0.08	
cor(Intercept,languagelatinhindi)	0.02
cor(languagehindi,languagelatinhindi)	0.23
cor(Intercept,Trialttypeincongruent)	-
0.17	
cor(languagehindi,Trialttypeincongruent)	0.15
cor(languagelatinhindi,Trialttypeincongruent)	0.01
cor(Intercept,languagehindi:Trialttypeincongruent)	-
0.06	
cor(languagehindi,languagehindi:Trialttypeincongruent)	-
0.03	
cor(languagelatinhindi,languagehindi:Trialttypeincongruent)	0.01
cor(Trialttypeincongruent,languagehindi:Trialttypeincongruent)	0.02
cor(Intercept,languagelatinhindi:Trialttypeincongruent)	-
0.14	
cor(languagehindi,languagelatinhindi:Trialttypeincongruent)	0.14
cor(languagelatinhindi,languagelatinhindi:Trialttypeincongruent)	-
0.04	
cor(Trialttypeincongruent,languagelatinhindi:Trialttypeincongruent)	0.10
cor(languagehindi:Trialttypeincongruent,languagelatinhindi:Trialttypeincongruent)	0.07
	Est.Error
sd(Intercept)	0.23
sd(languagehindi)	0.20
sd(languagelatinhindi)	0.22
sd(Trialttypeincongruent)	0.22
sd(languagehindi:Trialttypeincongruent)	0.15
sd(languagelatinhindi:Trialttypeincongruent)	0.23
cor(Intercept,languagehindi)	0.35
cor(Intercept,languagelatinhindi)	0.36
cor(languagehindi,languagelatinhindi)	0.36
cor(Intercept,Trialttypeincongruent)	0.35
cor(languagehindi,Trialttypeincongruent)	0.35
cor(languagelatinhindi,Trialttypeincongruent)	0.36
cor(Intercept,languagehindi:Trialttypeincongruent)	0.39
cor(languagehindi,languagehindi:Trialttypeincongruent)	0.39
cor(languagelatinhindi,languagehindi:Trialttypeincongruent)	0.38
cor(Trialttypeincongruent,languagehindi:Trialttypeincongruent)	0.38
cor(Intercept,languagelatinhindi:Trialttypeincongruent)	0.37
cor(languagehindi,languagelatinhindi:Trialttypeincongruent)	0.37
cor(languagelatinhindi,languagelatinhindi:Trialttypeincongruent)	0.37

cor(Trialtypeincongruent, languagelatinhindi: Trialtypeincongruent)	0.37
cor(languagehindi: Trialtypeincongruent, languagelatinhindi: Trialtypeincongruent)	0.39
	1-
95% CI	
sd(Intercept)	0.13
sd(languagehindi)	0.04
sd(languagelatinhindi)	0.04
sd(Trialtypeincongruent)	0.07
sd(languagehindi: Trialtypeincongruent)	0.00
sd(languagelatinhindi: Trialtypeincongruent)	0.01
cor(Intercept, languagehindi)	-
0.72	
cor(Intercept, languagelatinhindi)	-
0.66	
cor(languagehindi, languagelatinhindi)	-
0.53	
cor(Intercept, Trialtypeincongruent)	-
0.76	
cor(languagehindi, Trialtypeincongruent)	-
0.57	
cor(languagelatinhindi, Trialtypeincongruent)	-
0.67	
cor(Intercept, languagehindi: Trialtypeincongruent)	-
0.76	
cor(languagehindi, languagehindi: Trialtypeincongruent)	-
0.74	
cor(languagelatinhindi, languagehindi: Trialtypeincongruent)	-
0.70	
cor(Trialtypeincongruent, languagehindi: Trialtypeincongruent)	-
0.70	
cor(Intercept, languagelatinhindi: Trialtypeincongruent)	-
0.78	
cor(languagehindi, languagelatinhindi: Trialtypeincongruent)	-
0.63	
cor(languagelatinhindi, languagelatinhindi: Trialtypeincongruent)	-
0.72	
cor(Trialtypeincongruent, languagelatinhindi: Trialtypeincongruent)	-
0.61	
cor(languagehindi: Trialtypeincongruent, languagelatinhindi: Trialtypeincongruent)	-
0.68	
	u-
95% CI	
sd(Intercept)	0.96

sd(languagehindi)	0.82
sd(languagelatinhindi)	0.87
sd(Trialtypeincongruent)	0.89
sd(languagehindi:Trialtypeincongruent)	0.55
sd(languagelatinhindi:Trialtypeincongruent)	0.79
cor(Intercept,languagehindi)	0.61
cor(Intercept,languagelatinhindi)	0.69
cor(languagehindi,languagelatinhindi)	0.83
cor(Intercept,Trialtypeincongruent)	0.54
cor(languagehindi,Trialtypeincongruent)	0.74
cor(languagelatinhindi,Trialtypeincongruent)	0.68
cor(Intercept,languagehindi:Trialtypeincongruent)	0.69
cor(languagehindi,languagehindi:Trialtypeincongruent)	0.70
cor(languagelatinhindi,languagehindi:Trialtypeincongruent)	0.74
cor(Trialtypeincongruent,languagehindi:Trialtypeincongruent)	0.71
cor(Intercept,languagelatinhindi:Trialtypeincongruent)	0.59
cor(languagehindi,languagelatinhindi:Trialtypeincongruent)	0.78
cor(languagelatinhindi,languagelatinhindi:Trialtypeincongruent)	0.68
cor(Trialtypeincongruent,languagelatinhindi:Trialtypeincongruent)	0.77
cor(languagehindi:Trialtypeincongruent,languagelatinhindi:Trialtypeincongruent)	0.76
	Rhat
sd(Intercept)	1.00
sd(languagehindi)	1.00
sd(languagelatinhindi)	1.00
sd(Trialtypeincongruent)	1.00
sd(languagehindi:Trialtypeincongruent)	1.00
sd(languagelatinhindi:Trialtypeincongruent)	1.00
cor(Intercept,languagehindi)	1.00
cor(Intercept,languagelatinhindi)	1.00
cor(languagehindi,languagelatinhindi)	1.00
cor(Intercept,Trialtypeincongruent)	1.00
cor(languagehindi,Trialtypeincongruent)	1.00
cor(languagelatinhindi,Trialtypeincongruent)	1.00
cor(Intercept,languagehindi:Trialtypeincongruent)	1.00
cor(languagehindi,languagehindi:Trialtypeincongruent)	1.00
cor(languagelatinhindi,languagehindi:Trialtypeincongruent)	1.00
cor(Trialtypeincongruent,languagehindi:Trialtypeincongruent)	1.00
cor(Intercept,languagelatinhindi:Trialtypeincongruent)	1.00
cor(languagehindi,languagelatinhindi:Trialtypeincongruent)	1.00
cor(languagelatinhindi,languagelatinhindi:Trialtypeincongruent)	1.00
cor(Trialtypeincongruent,languagelatinhindi:Trialtypeincongruent)	1.00
cor(languagehindi:Trialtypeincongruent,languagelatinhindi:Trialtypeincongruent)	1.00
	Bulk_ESS

sd(Intercept)	1887
sd(languagehindi)	1490
sd(languagelatinhindi)	1738
sd(Trialtypeincongruent)	1378
sd(languagehindi:Trialtypeincongruent)	2235
sd(languagelatinhindi:Trialtypeincongruent)	1683
cor(Intercept,languagehindi)	4240
cor(Intercept,languagelatinhindi)	4404
cor(languagehindi,languagelatinhindi)	3629
cor(Intercept,Trialtypeincongruent)	3752
cor(languagehindi,Trialtypeincongruent)	3468
cor(languagelatinhindi,Trialtypeincongruent)	3308
cor(Intercept,languagehindi:Trialtypeincongruent)	6201
cor(languagehindi,languagehindi:Trialtypeincongruent)	4584
cor(languagelatinhindi,languagehindi:Trialtypeincongruent)	3562
cor(Trialtypeincongruent,languagehindi:Trialtypeincongruent)	3984
cor(Intercept,languagelatinhindi:Trialtypeincongruent)	4690
cor(languagehindi,languagelatinhindi:Trialtypeincongruent)	3679
cor(languagelatinhindi,languagelatinhindi:Trialtypeincongruent)	3576
cor(Trialtypeincongruent,languagelatinhindi:Trialtypeincongruent)	3067
cor(languagehindi:Trialtypeincongruent,languagelatinhindi:Trialtypeincongruent)	2731
	Tail_ESS
sd(Intercept)	2529
sd(languagehindi)	1709
sd(languagelatinhindi)	1567
sd(Trialtypeincongruent)	1231
sd(languagehindi:Trialtypeincongruent)	2371
sd(languagelatinhindi:Trialtypeincongruent)	2084
cor(Intercept,languagehindi)	3089
cor(Intercept,languagelatinhindi)	2941
cor(languagehindi,languagelatinhindi)	3424
cor(Intercept,Trialtypeincongruent)	3056
cor(languagehindi,Trialtypeincongruent)	3066
cor(languagelatinhindi,Trialtypeincongruent)	3104
cor(Intercept,languagehindi:Trialtypeincongruent)	2575
cor(languagehindi,languagehindi:Trialtypeincongruent)	3041
cor(languagelatinhindi,languagehindi:Trialtypeincongruent)	3521
cor(Trialtypeincongruent,languagehindi:Trialtypeincongruent)	3007
cor(Intercept,languagelatinhindi:Trialtypeincongruent)	3247
cor(languagehindi,languagelatinhindi:Trialtypeincongruent)	2746
cor(languagelatinhindi,languagelatinhindi:Trialtypeincongruent)	2857
cor(Trialtypeincongruent,languagelatinhindi:Trialtypeincongruent)	3281
cor(languagehindi:Trialtypeincongruent,languagelatinhindi:Trialtypeincongruent)	3292

Regression Coefficients:

	Estimate	Est.Error	l-95% CI	u-95% CI
Intercept	7.14	0.19	6.75	7.51
languagehindi	-0.15	0.16	-0.46	0.14
languagelatin hindi	-0.19	0.16	-0.51	0.13
Trialtypeincongruent	0.18	0.18	-0.18	0.56
languagehindi:Trialtypeincongruent	0.11	0.13	-0.14	0.36
languagelatin hindi:Trialtypeincongruent	-0.17	0.18	-0.51	0.16
	Rhat	Bulk_ESS	Tail_ESS	
Intercept	1.00	1731	2157	
languagehindi	1.00	1761	1551	
languagelatin hindi	1.00	2161	2109	
Trialtypeincongruent	1.00	1991	1954	
languagehindi:Trialtypeincongruent	1.00	3533	2645	
languagelatin hindi:Trialtypeincongruent	1.00	2573	1695	

Further Distributional Parameters:

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sigma	0.50	0.01	0.47	0.52	1.00	6667	2739

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS and Tail_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

Under regression coefficients, the l-95% and u-95% credible intervals (CrIs) all cross 0, thereby indicating the model cannot be confident about a correlation between the Hindi and Latin script conditions or the congruency conditions at 95% confidence.

```
fixef(ENHISTroopbrm, probs = c(0.1, 0.9)) |> round(2)
```

	Estimate	Est.Error	Q10	Q90
Intercept	7.14	0.19	6.94	7.34
languagehindi	-0.15	0.16	-0.32	0.01
languagelatin hindi	-0.19	0.16	-0.36	-0.01
Trialtypeincongruent	0.18	0.18	-0.02	0.38
languagehindi:Trialtypeincongruent	0.11	0.13	-0.04	0.26
languagelatin hindi:Trialtypeincongruent	-0.17	0.18	-0.36	0.02

At 80% CrIs, the upper and lower credible intervals cross 0 for all conditions except Latin Hindi congruent. This means the model is 80% confident that the RTs will decrease by -0.19

logged ms in the Latin Hindi congruent condition, but still not confident enough to identify a relationship between any of our other variables of interest. This finding is a stark contradiction with the hypothesis associated with our first research question.

```
fixef(ENHISTroopbrm, probs = c(0.15, 0.85)) |> round(2)
```

	Estimate	Est.Error	Q15	Q85
Intercept	7.14	0.19	6.99	7.30
languagehindi	-0.15	0.16	-0.28	-0.03
languagelatin hindi	-0.19	0.16	-0.32	-0.05
Trialtypeincongruent	0.18	0.18	0.02	0.33
languagehindi:Trialtypeincongruent	0.11	0.13	-0.01	0.23
languagelatin hindi:Trialtypeincongruent	-0.17	0.18	-0.31	-0.02

At 70% CrIs, the upper and lower estimate boundaries for Congruent Hindi, Incongruent English and Incongruent Hindi no longer cross 0, in addition to Congruent Latin Hindi as above in 80% CrIs. For every increase of 1 logged ms, the model is 70% confident that the RTs in the Congruent Hindi conditions will decrease by -0.28 to -0.03 logged ms, and between -0.32 and -0.05 logged ms for congruent Latin Hindi. It predicts that RTs in Incongruent English trials would increase between 0.02-0.33 logged ms for every increase in 1 logged ms at the intercept; this is the only condition where the model is 70% confident (or less) of an increase in reaction times.

To understand the effect of each of the language conditions and congruency conditions in greater depth, and unlog the values to plot them, we pulled the MCMC draws from the brms. We combined the intercept (English congruent) with the interactions of interest (Hindi congruent, Latin Hindi congruent, English incongruent, Hindi incongruent, Latin Hindi incongruent, Hindi incongruent interaction and latin Hindi incongruent interaction) to establish the posterior estimates for these 6 conditions.

```
#draws so we can pull summary measures
stroopbrm_draws <- as_draws_df(ENHISTroopbrm)
stroopbrm_draws
```

```
# A draws_df: 1000 iterations, 4 chains, and 61 variables
  b_Intercept b_languagehindi b_languagelatin hindi b_Trialtypeincongruent
1          7.3          -0.193          -0.297              0.118
2          7.3          -0.185          -0.242             -0.043
3          7.2          -0.054          -0.030              0.111
4          7.2          -0.238          -0.243              0.136
5          7.2          -0.312          -0.341              0.096
6          7.2          -0.388          -0.293              0.133
```

7	7.2	-0.055	-0.104	0.392
8	7.1	0.064	-0.112	0.318
9	7.2	-0.109	-0.208	0.287
10	7.2	-0.098	-0.021	0.114

b_languagehindi:Trialttypeincongruent

1	0.2083
2	0.1221
3	0.0096
4	0.1913
5	0.2840
6	0.3306
7	-0.0817
8	-0.0690
9	0.0217
10	0.0285

b_languagelatin hindi:Trialttypeincongruent sd_participant_ID__Intercept

1	-0.024	0.43
2	-0.193	0.28
3	-0.352	0.35
4	0.041	0.16
5	0.074	0.18
6	0.074	0.13
7	-0.385	0.20
8	-0.293	0.24
9	-0.334	0.13
10	-0.229	0.11

sd_participant_ID__languagehindi

1	0.094
2	0.140
3	0.402
4	0.164
5	0.121
6	0.246
7	0.263
8	0.223
9	0.213
10	0.231

... with 3990 more draws, and 53 more variables
... hidden reserved variables {'.chain', '.iteration', '.draw'}

```
#Mutating the draws to get the intercepts for all six conditions of interest
```

```

stroopbrm_draws2 <- stroopbrm_draws |>
  mutate(
    english_congruent = b_Intercept,
    hindi_congruent = b_Intercept + b_languagehindi,
    latin_hindi_congruent = b_Intercept + b_languagelatin_hindi,
    english_incongruent = b_Intercept + b_Trialtypeincongruent,
    hindi_incongruent = b_Intercept + b_languagehindi + b_Trialtypeincongruent
    + `b_languagehindi:Trialtypeincongruent`,
    latin_hindi_incongruent = b_Intercept + b_languagelatin_hindi
    + b_Trialtypeincongruent + `b_languagelatin_hindi:Trialtypeincongruent`
  )

```

#Selecting the relevant columns and pulling them into a new tibble for clarity.

```

stroopbrm_draws3 <- stroopbrm_draws2 |>
  select(english_congruent, hindi_congruent, latin_hindi_congruent,
         english_incongruent, hindi_incongruent, latin_hindi_incongruent)

```

Warning: Dropping 'draws_df' class as required metadata was removed.

The Lognormal regression model automatically logs the RTs to approximate them to a Gaussian distribution; however, for ease of interpreting the summary estimates, we can unlog them to retrieve the estimated RTs in ms from the draws.

```

# unlogging the RTs; mutate is used to create new columns

stroopdrawsunlog <- stroopbrm_draws3 |>
  mutate(
    english_congruent_log = english_congruent,
    hindi_congruent_log = hindi_congruent,
    latin_hindi_congruent_log = latin_hindi_congruent,
    english_incongruent_log = english_incongruent,
    hindi_incongruent_log = hindi_incongruent,
    latin_hindi_incongruent_log = latin_hindi_incongruent,
    english_congruent_unlog = exp(english_congruent),
    hindi_congruent_unlog = exp(hindi_congruent),
    latin_hindi_congruent_unlog = exp(latin_hindi_congruent),
    english_incongruent_unlog = exp(english_incongruent),
    hindi_incongruent_unlog = exp(hindi_incongruent),
    latin_hindi_incongruent_unlog = exp(latin_hindi_incongruent)
  )

```

As the model did not have confidence in a relationship between conditions at 95% intervals, we can explore the summary measures of the regression coefficients (logged) at 85, 75, and 66% confidence intervals.

```
#pivot longer LOGGED VALUES
stroopdrawslog_long <- stroopdrawsunlog |>
  select(english_congruent_log, english_incongruent_log,
         hindi_congruent_log, hindi_incongruent_log, latin_hindi_congruent_log,
         latin_hindi_incongruent_log) |>
  pivot_longer(everything(), names_to = "condition", values_to = "RT_ms")

#summarise pivot longer
stroopdrawslog_long_tab <- stroopdrawslog_long |>
  group_by(condition) |>
  summarise(
    mean = mean(RT_ms), sd = sd(RT_ms),
    '95%' = paste0("[", paste(quantile(RT_ms, c(0.025, 0.975)) |> round(2),
                                collapse = ", "), "]"),
    '85%' = paste0("[", paste(quantile(RT_ms, c(0.075, 0.925)) |> round(2),
                                collapse = ", "), "]"),
    '75%' = paste0("[", paste(quantile(RT_ms, c(0.125, 0.875)) |> round(2),
                                collapse = ", "), "]"),
    '66%' = paste0("[", paste(quantile(RT_ms, c(0.17, 0.83)) |> round(2),
                                collapse = ", "), "]")
  )

stroopdrawslog_long_tab
```

```
# A tibble: 6 x 7
  condition          mean    sd `95%`      `85%`      `75%`      `66%`
  <chr>          <dbl> <dbl> <chr>      <chr>      <chr>      <chr>
1 english_congruent_log  7.14 0.188 [6.75, 7.51] [6.9, 7.38] [6.96~ [7, ~
2 english_incongruent_log 7.32 0.240 [6.82, 7.79] [7.02, 7.63] [7.1,~ [7.1~
3 hindi_congruent_log    6.99 0.227 [6.51, 7.44] [6.7, 7.27] [6.77~ [6.8~
4 hindi_incongruent_log  7.28 0.286 [6.68, 7.87] [6.9, 7.64] [7, 7~ [7.0~
5 latin_hindi_congruent_log 6.95 0.238 [6.47, 7.42] [6.64, 7.26] [6.72~ [6.7~
6 latin_hindi_incongruent_log 6.96 0.313 [6.3, 7.58] [6.54, 7.37] [6.65~ [6.7~
```

We can now do the same with the unlogged values to establish the summary measures of the regression coefficients (unlogged) at 85, 75, and 66% confidence intervals.

```

#pivot longer UNLOGGED VALUES
stroopdrawsunlog_long <- stroopdrawsunlog |>
  select(english_congruent_unlog, english_incongruent_unlog,
         hindi_congruent_unlog, hindi_incongruent_unlog, latin_hindi_congruent_unlog, latin_hindi_incongruent_unlog,
         pivot_longer(everything(), names_to = "condition", values_to = "RT_ms"))

#summarise pivot longer
stroopdrawsunlog_long_tab <- stroopdrawsunlog_long |>
  group_by(condition) |>
  summarise(
    mean = mean(RT_ms), sd = sd(RT_ms),
    '95%' = paste0("[", paste(quantile(RT_ms, c(0.025, 0.975)) |> round(2),
                                collapse = ", "), "]"),
    '85%' = paste0("[", paste(quantile(RT_ms, c(0.075, 0.925)) |> round(2),
                                collapse = ", "), "]"),
    '75%' = paste0("[", paste(quantile(RT_ms, c(0.125, 0.875)) |> round(2),
                                collapse = ", "), "]"),
    '66%' = paste0("[", paste(quantile(RT_ms, c(0.17, 0.83)) |> round(2),
                                collapse = ", "), "]")
  )

stroopdrawsunlog_long_tab

```

```

# A tibble: 6 x 7
  condition          mean    sd `95%`          `85%` `75%` `66%`
  <chr>          <dbl> <dbl> <chr>          <chr> <chr> <chr>
1 english_congruent_unlog 1289. 298. [851.78, 1819.11] [994.~ [105~ [109~
2 english_incongruent_unlog 1559. 448. [912.95, 2424.92] [1118~ [120~ [125~
3 hindi_congruent_unlog   1114. 290. [672.14, 1709.76] [815.~ [871~ [913~
4 hindi_incongruent_unlog  1509. 514. [799.58, 2615.73] [993.~ [109~ [115~
5 latin_hindi_congruent_unlog 1076. 305. [648.47, 1671.31] [765.~ [825~ [861~
6 latin_hindi_incongruent_unlog 1108. 390. [544.2, 1953.29] [689.~ [769~ [819~

```

The table above shows at what confidence values we can expect the posterior mean RTs to fall under for the 6 conditions of interest. In the congruent conditions, English has the highest posterior mean RTs at 1288.767ms, followed by Hindi at 1114.456ms and lastly Latin Hindi at 1075.772ms. This mimics the pattern established by the raw values above.

In the incongruent conditions, the pattern is the same: English trials had the longest RTs, followed by Hindi and then Latin script Hindi. This indicates that, even mitigating for participant variability, participants are predicted to have the fastest RTs reading Latin script Hindi.

Now we have the posterior summary measures from the regression model, we can plot the data to compare how the data is distributed.

```
#| label: plot: conditional effects log
```

```
conditional_effects(ENHIstroopbrm)
```

```
Ignoring unknown labels:
```

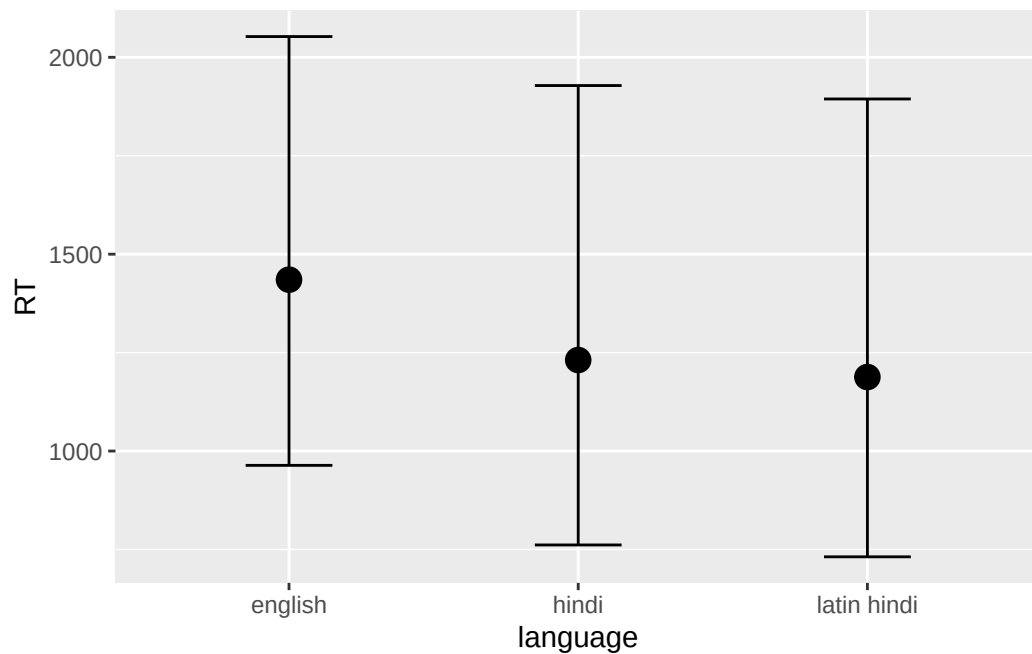
```
* fill : "NA"
```

```
* colour : "NA"
```

```
Ignoring unknown labels:
```

```
* fill : "NA"
```

```
* colour : "NA"
```



```
Ignoring unknown labels:
```

```
* fill : "NA"
```

```
* colour : "NA"
```

```
Ignoring unknown labels:
```

```
* fill : "NA"
```

```
* colour : "NA"
```

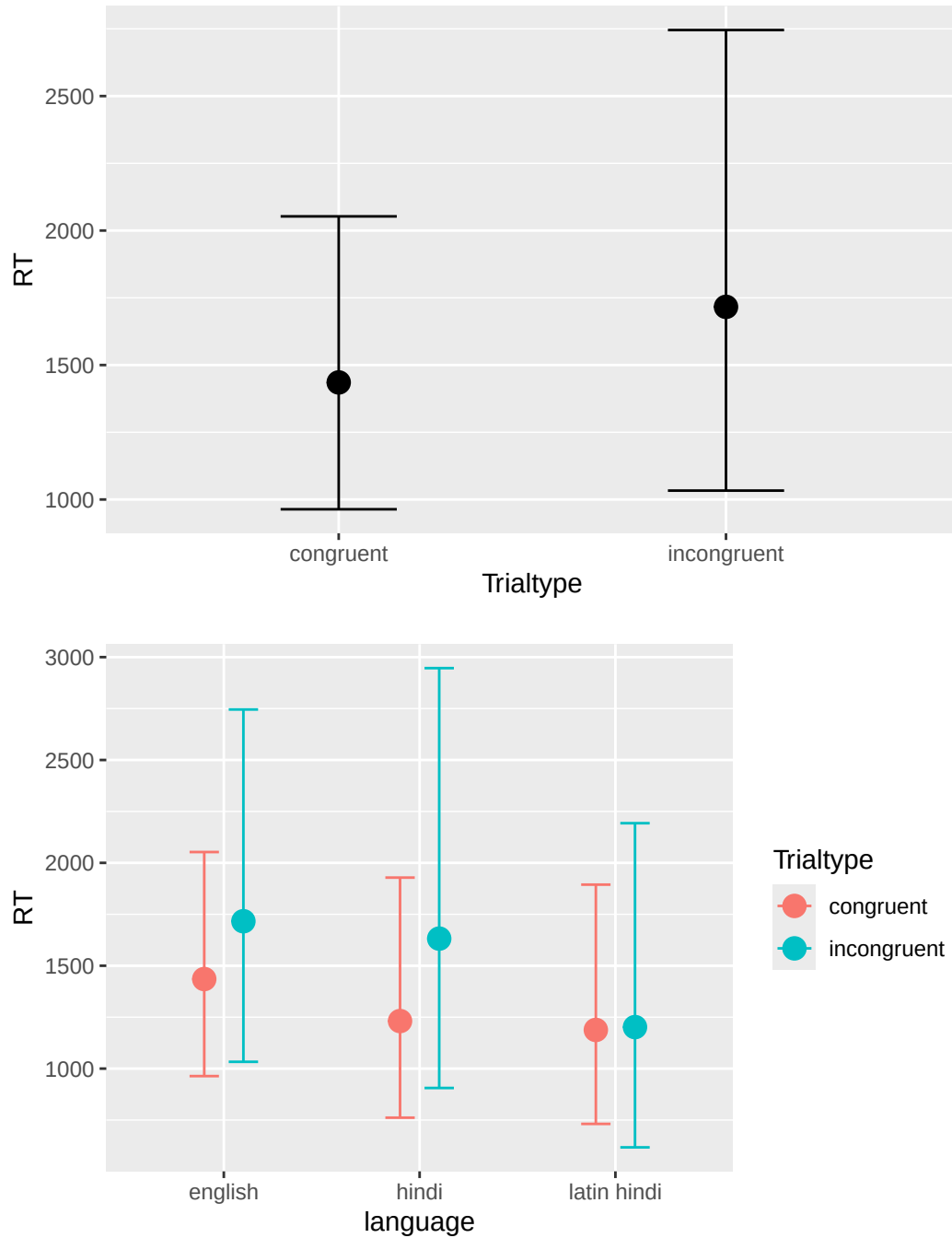


Figure 3 shows a conditional effects whisker plot. Conditional effects of the brms model indicate a pattern of greater variance in RTs across all incongruent conditions. Posterior mean RTs were higher in English in both congruency conditions compared the equivalent conditions in Latin Hindi or Devanagari Hindi. This indicates a higher probability of the Stroop effect in

English incongruent trials, compared to Hindi and Latin Hindi incongruent trials.

```
#| label: plot: half-eye unlog

stroopdrawsunlog_long |>
  ggplot(aes(x = RT_ms, y = condition, fill = condition)) +
  stat_halfeye() +
  coord_cartesian(xlim = c(0, 3000))+
  labs(
    x = "Predicted RTs", y = "Conditions",
    title = "Figure 4: half-eye plot showing the distribution of RTs across
    difference language and congruency conditions",
    caption = "Point = mean, thick bar = 66% CrI, thin bar = 95% CrI."
  )
```

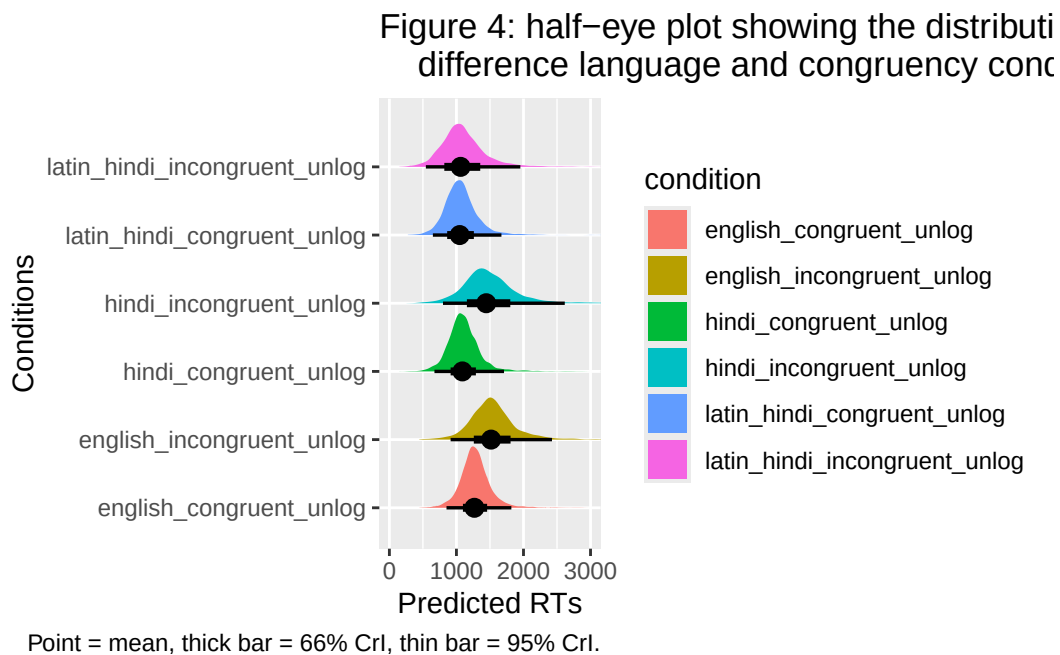


Figure 4 above is a half-eye plot comparing the posterior means, 66% CrIs, 95% CrIs, and probability distribution of RTs across all 3 language conditions and congruency conditions. This plot shows that, even controlling for participant variability, there is considerably more variance in the English and Hindi incongruent predicted RTs. In contrast to the raw data plot above in Section 3.4, this plot indicates a stronger pattern of Stroop effect in English and Devanagari Hindi with higher mean RTs, but not in Latin Script Hindi. The greatest difference in congruency within language conditions appears to be Congruent and Incongruent Hindi; however, this degree of difference is marginal compared to the difference between English

congruency conditions. It is also important to stress the degree of overlap between the posterior means and CrIs of all 6 conditions plotted here, reinforcing that the model cannot be confident in a correlation between the conditions and an effect on RTs.

5 5. Discussion

This study set out to investigate the effect of orthography on non-selective activation in Hindi-English bilinguals with a Stroop task which manipulated script in Hindi while holding the phonology of the language constant.

Our first research question asked whether Hindi-English bilinguals have longer reaction times, and therefore a greater Stroop effect, reading Latin script Hindi words versus reading Devanagari Hindi words. We hypothesised that reaction times in incongruent Latin Script trials would be greater compared to incongruent Devanagari Hindi trials, indicating a greater Stroop effect. However, our regression model not able to confidently identify a combined relationship between language and incongruence at 95% or 80% CrIs, and thus our regression results are inconclusive for this research question. Surprisingly, the pattern indicated by the raw data and the summary measures would suggest that Latin Hindi reaction times for incongruent and congruent conditions were similar and faster than English or Devanagari Hindi conditions. This is an intriguing result worthy of further exploration in a future study.

Secondly, we asked which language condition had the greatest degree of interference between congruent and incongruent trials across language conditions. The BIA+ model hypothesised that the greatest magnitude of interference would be found in the Latin Script Hindi model as the script would not be an effective suppression cue and the phonology of the words would be the same, causing significant interference. Once again, our regression model was not able to confidently identify an effect between these conditions at 95% confidence intervals, so we must conclude that the results also remain inconclusive.

Nonetheless, these initial results from the raw data would contradict the predictions of the BIA+ model, which claimed that script differences with similar (or identical) phonology would create greater interference during non-selective activation for bilinguals.

5.1 5.1 Limitations

We identified a small number of limitations in the present study which may have emerged due to the small scope of the project or as a byproduct of the methodological design. The first was the sample size; a larger group of participants were recruited but unfortunately not included in the study as they failed to respond in time. The size of this sample, combined with the varying effects fitted in the regression model, decreases the model's confidence in any relationships between the results. Researchers seeking to replicate the present study should address this as a primary concern.

Additionally, the order of trials in the Testable experiment may have contributed to the pattern of effect seen in the language condition. RTs in the English trials - which came first in the experimental design - were longer than that of Hindi which followed, or Latin Hindi which were lower still than Hindi. It is possible that participants' overall competency and reaction times in the Stroop task improved with experience (i.e., greater familiarity with the colour buttons on the keyboard) which mitigated the effect of language on the Stroop task. Possible solutions to this include counterbalancing the order of trials for participants to mitigate the effect of practice, or replicating this with oral responses instead of keyboard responses. Additionally, it is worth considering that the participants' reading proficiency scores varied considerably compared to their English scores. A stronger reading proficiency in English would plausibly skew the results, and the Latin script combined with the Hindi phonology would provide a much stronger cue. It would be therefore be of great interest to conduct the experiment again with a similar cohort of participants and collecting oral data rather than push data, as this could have clearly impacted the Stroop effect seen here.

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